

IMAGE-TO-BIOMASS – PREDICTION USING COMPUTER VISION AND DEEP LEARNING

Final Documentation Report

Project Report Submitted for Internship Project Submission

Submitted By

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Internship Role: Intern (AI/ML)

Project Title: Image-to-Biomass Prediction System

Technology Stack:

Python | Computer Vision | Deep Learning | PyTorch | OpenCV | Streamlit | Scikit-Learn
| Pandas | NumPy

Project Description:

An AI-powered precision agriculture solution developed to estimate pasture biomass directly from RGB images. The project leverages deep learning (EfficientNet) for visual feature extraction and computer vision (HSV segmentation, pseudo-depth estimation) to generate precise numerical biomass predictions in kg/ha. A premium Streamlit dashboard provides interactive assessments, visual topographic mapping, and Grad-CAM interpretability for data-driven farming decisions.

Submitted To

Company/Organization Name: Data Vidwan

Mentor/Manager Name: Meet Mistry

Date of Submission: 10/07/2026

1. Project Objective

The objective of this project is to develop an AI-driven agricultural decision support system that helps farmers and agronomists evaluate their pasture biomass yields automatically, replacing the traditional labor-intensive 'clipping and weighing' method.

The system uses deep learning-based visual feature extraction (EfficientNet-B0), traditional computer vision for semantic segmentation, and environmental metadata processing to predict dry biomass yields.

The project includes:

- Dataset Preprocessing and Augmentation
- Semantic Segmentation Model Implementation (DeepLabV3+ / HSV)
- Biomass Regression Model Training (EfficientNet-B0)
- Zero-Download Computer Vision Depth Estimation
- Grad-CAM Heatmap Interpretability
- Interactive, Premium Streamlit Dashboard Deployment

2. Requirement Coverage

Requirement	Status	Project File / Output
Semantic Segmentation	Completed	src/models/segmentation_model.py / dashboard/app.py
Biomass Prediction Model	Completed	models/best_model.pth
Depth Estimation	Completed	dashboard/app.py (CV Pseudo-Depth)
Visual Heatmaps	Completed	src/visualization/gradcam.py
Streamlit Dashboard	Completed	dashboard/app.py

3. Dataset

The project utilizes a rich agricultural dataset for training and assessment:

Pasture Biomass RGB Dataset

- Multi-category biomass data including Green Dry Matter, Dead Material, and Clover.
- Includes environmental metadata (NDVI, Canopy Height, Species, Geographic State).

4. Work Completed

Completed an end-to-end AI/ML workflow including:

- Processed and standardized agricultural RGB images and CSV metadata.
- Engineered an EfficientNet-B0 backbone for visual feature extraction.
- Implemented metadata scaling and encoding for dual-input fusion.
- Developed a robust regression model to predict 5 distinct biomass categories in kg/ha.
- Built a premium, interactive Streamlit dashboard (Precision Pasture AI).
- Integrated Grad-CAM for visual model explainability.
- Designed Zero-Download OpenCV algorithms for semantic segmentation and topographic depth approximation.

5. Methodology

5.1 Data Acquisition & Preprocessing

Data quality was maintained by applying standard PyTorch transformations (resizing, normalization) to all incoming RGB images. Environmental metadata was normalized using Scikit-Learn's StandardScaler and categorical features were Label Encoded.

5.2 Feature Extraction

- Visual: Spatial feature mapping via EfficientNet-B0 convolutional layers.
- Metadata: Linear layer processing of continuous and categorical environmental inputs.
- Subjective: Computer vision estimations for canopy height and vegetation coverage percentages.

6. Model Development

Multiple approaches were considered for the distinct modalities:

Architectures Evaluated:

- EfficientNet-B0 (Transfer Learning)
- DeepLabV3_ResNet50 (Semantic Segmentation)
- MiDaS (Monocular Depth Estimation)

7. Model Selection

Selected Models:

1. Regression: EfficientNet-B0. Provided the best generalization on complex field textures with low parameter overhead.
2. Segmentation & Depth: OpenCV HSV color mapping and Gaussian pseudo-depth were selected for the production web app to ensure zero-latency inference and respect server memory limits, replacing DeepLab and MiDaS for the final deployment.

8. Streamlit Application

The Streamlit application is implemented in:

dashboard/app.py

The dashboard contains:

Assessment Flow Page (app.py):

- Step 1: Image Upload for Top-Down RGB Field Image.
- Step 2: Configuration of Environmental Metadata Sidebar.
- Step 3: Fused Readout generating Biomass Yields in kg/ha.
- Step 4: Visual Intelligence suite (Semantic Mask, Depth Map, Grad-CAM Heatmap).

Metrics & About Pages:

- Interactive Training Loss vs Validation Loss charts.
- Evaluation plots (Actual vs Predicted, Feature Correlation).
- System Architecture and project documentation.

9. Key Findings

- Fusing environmental metadata (NDVI, Height) with visual features significantly outperformed single-modality image evaluations.
- The inclusion of geographical state and species stabilizes highly-variant visual predictions.
- Traditional CV algorithms (HSV, Grayscale) provided faster and equally actionable visual insights compared to heavy deep learning models (MiDaS) for the web deployment.

10. Recommendations

- Utilize the Grad-CAM heatmaps to verify the model is looking at vegetation and not anomalies (e.g., equipment in the field).
- Encourage users to input highly accurate NDVI and Canopy Height metadata to ensure optimal precision.

11. Achievements

Successfully completed:

- Multimodal feature extraction (Image + Metadata).
- Independent Deep Learning model training and validation.
- Implementation of a dual-input PyTorch architecture.
- End-to-End Streamlit Dashboard deployment with an advanced, stylized 'Precision Pasture' UI.
- Real-time computer vision processing and Grad-CAM generation.

12. Limitations

- The model relies heavily on a top-down orthogonal angle; severely angled photos may reduce accuracy.
- The free-tier deployment requires CV approximation for depth rather than true stereoscopic or neural depth mapping due to RAM constraints.

13. Final Submission Contents

- Documentation Report (docs/Final_Documentation_Report.docx)
- Machine Learning Models & Scalers (models/)
- Streamlit Dashboard (dashboard/app.py, dashboard/pages/)
- Visual Utilities (src/visualization/gradcam.py)
- Problem Statement (PROBLEM_STATEMENT.md)

14. How to Run

```
pip install -r requirements.txt
```

```
python -m streamlit run dashboard/app.py
```

15. Final Conclusion

The Precision Pasture AI platform successfully delivers an end-to-end multimodal machine learning solution for agricultural biomass assessment.

By seamlessly integrating deep convolutional image analysis with critical environmental metadata, the system minimizes the blind spots of single-modality AI. Delivered via a highly interactive and beautifully stylized Streamlit dashboard, this project provides farmers and agronomists with immediate, data-driven insights and a highly scalable alternative to manual field harvesting.